

Topic: Microinstruction Sequencing and Execution

Q. Explain the concepts of Microinstruction Sequencing and Execution in a Microprogrammed Control Unit. What is Control Memory?

> **Definition to Remember**

Microinstruction Sequencing & Execution

- It is a fast, Read-Only Memory (ROM) located inside the CPU, completely separate from main system memory.

2. Microinstruction Execution

A microinstruction is basically a control word. The bits in this word directly translate to physical control lines in the CPU.

3. Microinstruction Sequencing

Sequencing is the process of determining the address of the *next* microinstruction to fetch from Control Memory. The hardware unit responsible is the **Sequencer**.

4. Workflow Loop

1. Map Machine Instruction Opcode to Control Memory starting address.

> **Must-Write Points (for 10 marks)**

> **Quick Recall**

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